



SRN Solar PV – Khaldiya

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ENG208 : Introduction to Sustainable Energy

Spring 2025/2026

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SRN SOLAR PV KHALDIYA DISTRICT

Sustainable Energy System Design

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1. INTRODUCTION

What would happen if the electricity went out suddenly in Kuwait, especially given the recent challenges? This has caused widespread fear, leading many to consider buying generators. In contrast, many developed neighboring countries were better prepared for such situations. Our investigation revealed that they have heavily relied on renewable energy, particularly solar panels. This inspired the idea for our project, with a primary focus on the Khaldiya district.

Kuwait needs to rely on renewable energy sources because it has depended heavily on fossil fuels, which have negatively impacted the environment. Unlike fossil fuels, renewable energy sources are sustainable and naturally replenished, such as sunlight and wind. Kuwait relies primarily on oil and natural gas for electricity generation. Due to the extremely hot weather, especially in the summer, residents use air conditioning extensively, creating a significant demand for electricity. Furthermore, factories in Kuwait require substantial amounts of electricity, and unfortunately, they rely on fossil fuels to operate. Therefore, Kuwait must shift toward cleaner energy sources such as solar energy, which is cleaner to use, in order to reduce pollution and meet future energy needs.

Solar energy is very suitable for Kuwait because it has high solar radiation throughout the year, which means that using solar panels is the best way to efficiently produce a large amount of electricity. Solar energy is also clean and helps reduce pollution and dependence on gas and oil. This project focuses on designing a photovoltaic (PV) solar energy system for the Al-Khalidiyah area. The primary objective is to provide a sustainable and reliable source of electricity and reduce reliance on fossil fuels. This study includes estimating the area's electricity demand, calculating the required solar system capacity, and assessing the environmental, social, and economic impacts. Additionally, the project explores how the use of renewable energy can contribute to improved sustainability and development within the area.

2. DISTRICT DESCRIPTION

Al-Khalidiyah is a well-developed residential area located in Kuwait City Governorate. The area is characterized by a blend of residential buildings, educational institutions, and local services, including educational institutions such as Abdullah Al-Salem University. The population of the area is estimated at approximately 19,000 to 20,000 based on available demographic data (Central Statistical Bureau, n.d.). As a developed urban area, Al-Khalidiyah consumes a significant amount of electricity, increasing the need for a reliable and, importantly, clean energy source. The primary reason for this increased electricity demand is primarily due to households, commercial activities, and the intensive use of air conditioning systems, especially given the extremely hot summer months (International Energy Agency, n.d.). Due to the increased energy demand in the area, the implementation of renewable energy systems, such as solar photovoltaic systems, is very suitable for the district.

3. ENERGY DEMAND ESTIMATION

The annual electricity consumption for the Al-Khalidiyah area was estimated based on the population. Since there is no precise data on the area's electricity consumption rate, it was assumed that the average annual electricity consumption in Kuwait is 15,000 kWh/person/year. This reflects the high energy consumption in the country, due to the hot and harsh climate, which necessitates the intensive use of air conditioning systems (Course Project Guidelines, 2025). As previously mentioned, the population of the district is estimated at 19,300.

Based on these assumptions and information, the total annual electricity demand was calculated by multiplying the region's population by per capita consumption. Using this formula:

$$\begin{aligned}\text{Annual demand} &= \text{Population} \times \text{Consumption per capita} \\ &= 19,300 \times 15,000 \\ &= 289,500,000 \text{ kWh/year, equivalent to } 289.5 \text{ GWh/year.}\end{aligned}$$

This estimated electricity demand for the region will be used in our project as the basis for determining the capacity of the proposed solar photovoltaic (PV) system.

4. RENEWABLE ENERGY SYSTEM DESIGN

Khaldiya is a densely populated urban residential area in Kuwait, which is known for its high solar potential due to strong sunlight throughout the year. Implementing solar photovoltaic system deems perfect in order to harvest and harness solar energy that Kuwait yields. There are several ways to maximize efficiency such as adding Perovskites (M. Mahmoud 2025) . Firstly, the solar panel type has to be mono-crystalline because it absorbs more energy compared to polycrystalline. Secondly, 30% of energy is lost due to sunlight being reflected off the solar panel, therefore, investing in anti-reflection coating with bifocals panels may increase cell efficiency. Thirdly, integrating ground-mounting solar arrays where trackers allow panels to follow sunlight may significantly increase solar efficiency. Finally, Khaldiya has twelve schools, fifteen mosques, cooperative societies and government office in addition to residential areas, covering their rooftops with solar panels should provide 88 MW worth of capacity. In addition to solar carports, 1.24 kilometers square is estimated whereas the required is 1.15 kilometers squared which is more than sufficient. According to NREL (2023) 128MW PV system should generate 218GWh annually, meeting 50% of the districts demands due to 5.5kWh/ meter square/ day of solar irradiance (Al-Shammari & Al-Mahdi,2022).



Figure1. The figure above shows SRN Solar PV logo

5. SYSTEM CAPACITY CALCULATION RENEWABLE ENERGY SYSTEM DESIGN: -

In a tropical sunny climate such as Khaldiya, a 1 kW solar photovoltaic system approximately after losses such as shading and seasons changes. In order to meet Khaldiya's annual renewable energy demands of 144,750,000 kWh which represents 50% of total energy consumption, 85MW of installed solar capacity is required. With simple calculations, 10,000 meters square per MW for ground-mounted fixed-tilt arrays translates to 850,000 meters square which can be distributed across rooftops, parking lot shade structures with integrated photovoltaic panels.

6. ENVIRONMENTAL IMPACT

By generating 144,750,000 kWh of clean electricity annually through the proposed 85MW solar system would avoid approximately 72,375 metric tons of carbon dioxide each year,(U.S EPA, 2023) which is equivalent to planting 1.2 million trees and allowing them to grow for a decade. Displacing fossil fueled generations cuts harmful co-pollutants such as sulfur dioxides and nitrogen oxides and fine particulate matter (National Research Council, 2010). When cutting back on fossil fuels, pollution is reduced significantly which leads to significant reduction in hospital admissions for respiratory diseases. Overall, it has been proven that reducing fossil fuel use provides many benefits that include lower rates of asthma and other respiratory diseases, improved local air quality and greater energy independence for the district.(Munoz et al., 2021)

7. SOCIAL IMPACT

The SRN Solar PV Project in Khaldiya is expected to provide several social benefits to the local community. One of its main advantages is the creation of jobs during the construction, installation, operation, and maintenance phases. Proper installation and long-term operation of the solar energy system will require engineers, technicians, project managers, and support staff. This can lead to additional employment opportunities and the development of technical skills in the renewable energy sector.

Another advantage is increasing public awareness of renewable energy and sustainability. Residents may become more interested in adopting sustainable practices such as energy saving and rooftop solar systems after detecting the usage of solar panels and clean energy technologies in their region. This can help to build a stronger environmental culture in Khaldiya.

The SRN Solar PV Project may also help improve public health by reducing air pollution generated by traditional fossil-fuel power generation. Lower emissions can enhance air quality and help to prevent respiratory diseases and other pollution-related health issues. Furthermore, the SRN Solar PV Project can improve energy dependability by supplying electricity to the local grid, particularly during peak summer demand in Kuwait.

However, some transitory issues may arise during the project's early stages. Residents may be inconvenienced by construction noise, traffic disruptions, or other factors. Some neighbourhood members may first oppose changes or be unfamiliar with renewable energy systems. Despite these limitations, the long-term social advantages are predicted to surpass the immediate difficulties.

8. ECONOMIC IMPACT

The SRN Solar PV Project in Khaldiya requires an essential initial investment, estimated at around 59.5 million KD for an 85 MW system (calculated from $85\text{MW} * 700,000 \text{ KD/MW}$). This cost covers solar panels, inverters, electrical equipment, installation, grid connection, infrastructure development, and maintenance planning. Despite the high initial cost, solar energy projects are viewed as long-term investments with significant potential returns.

One of the most important economic advantages is the reduction in long-term electricity generation costs. Because sunshine is a free and abundant resource in Kuwait, post-installation operational costs are significantly lower than those of typical fossil fuel power plants. This can help to lessen the long-term financial burden of electricity production while also promoting national energy savings. The project may also lessen reliance on fossil fuels such as oil and natural gas. This enables important hydrocarbon resources to be kept for export or reallocated to higher-value businesses. Furthermore, investment in renewable energy promotes economic diversity, which is a critical goal for Kuwait's future economy. Another economic benefit is the creation of local job opportunities in the engineering sector, especially for AASU graduates and majors such as Energy Systems Engineering, Environmental and Sustainability Engineering. These majors can directly contribute to the planning, operation, and development of many projects.

According to Dr. Morsi Ahmed, one of the major challenges facing renewable energy projects in the Gulf region is the high cost of infrastructure transformation, since most existing systems were originally designed for traditional oil and gas energy sources (M. Ahmed, personal communication, November 3, 2025). This highlights the importance of strategic planning and phased investment when implementing the SRN Solar PV Project in Khaldiya. Although the payback period may take several years, the long-term economic benefits make the project financially beneficial and valuable.

9. HUMAN DEVELOPMENT INDEX (HDI) IMPACT

9.1 Life expectancy index

The Life Expectancy Index assesses the health aspect of human growth. It is calculated with the formula:

$$LEI = (LE - 20) / (85 - 20)$$

LE denotes life expectancy at birth. Khaldiya's life expectancy is predicted to be 78 years, which is similar to Kuwait's national average due to the district's convenient access to healthcare facilities and modern living circumstances. Therefore:

$$LEI = (78 - 20) / 65 = 58 / 65 = 0.892$$

This score suggests that the district has good health and living conditions. Furthermore, the SRN Solar PV Project in Khaldiya may enhance health by lowering air pollution and greenhouse gas emissions associated with conventional electricity generation.

9.2 Educational Index.

The Education Index measures access to education and learning opportunities. It is based on two measures: mean years of schooling (MYS) and expected years of schooling (EYS).

First, the mean years of schooling index is calculated as follows:

$$MYSI = MYS / 15.$$

Khaldiya's projected MYS is 11.5 years.

$$MYSI = 11.5 / 15 = 0.767$$

Second, the Expected Year of Schooling Index is:

$$EYSI = EYS/18$$

The anticipated EYS is seventeen years:

$$EYSI = 17 / 18 = 0.944$$

The Education Index is calculated as follows:

$$EI \text{ equals } (MYSI + EYSI) / 2$$

$$EI = (0.767 + 0.944) / 2 = 0.856$$

This indicates a high educational level in Khaldiya. The SRN Solar PV Project can provide reliable electricity to schools, universities, laboratories, and digital learning systems.

9.3 INCOME INDEX & FINAL HDI

The Income Index calculates the standard of living using Gross National Income (GNI) per capita. The standard HDI technique generally expresses it in USD, but for this project, the value can also be reported in Kuwaiti Dinar (KWD) to ensure local relevance. Khaldiya's projected GNI per capita is at 16,000 KWD, or about \$52,000 USD.

The Income Index is computed with the logarithmic formula:

$$\text{Income Index (II)} = [\ln(\text{GNIpc}) - \ln(100)] \text{ divided by } [\ln(75000) - \ln(100)].$$

Using an equal international value:

$$\text{II} = [\ln(52000) - \ln(100)] \text{ divided by } [\ln(75000) - \ln(100)]$$

$$\text{II} = (10.859 - 4.605) / (11.225 - 4.605)$$

$$\text{II} = 6.254 / 6.620 = 0.945$$

After calculating the three indices, the final HDI is determined using the geometric mean:

$$\text{HDI} = (\text{LEI} \times \text{EI} \times \text{II})^{(1/3)}.$$

$$\text{HDI} = (0.892 \times 0.856 \times 0.945)^{(1/3)}$$

$$\text{HDI} = (0.721)^{(1/3)} = 0.897$$

This estimated value reflects a very high degree of human development in Khaldiya. The SRN Solar PV Project has the potential to grow this value over time by improving health, providing improved education support, and expanding economic prospects. The SRN Solar PV Project in Khaldiya has the potential to boost the district's human development index. It may benefit health by lowering air pollution and emissions from traditional power generating. It can also help with education by providing reliable electricity to schools, laboratories, and digital learning systems. Furthermore, the project may generate job opportunities and lower long-term energy prices, hence raising income levels and living standards. Overall, the project has the potential to improve the quality of life and promote long-term growth in Khaldiya.

10. CONCLUSION

In conclusion, this project demonstrated that the use of photovoltaic (PV) solar energy is a highly effective and suitable solution for the Al-Khaldiya area. Based on the estimated annual electricity demand of approximately 289.5 GWh, these panels were designed to provide a significant portion of the residents' energy needs. Furthermore, the use of renewable energy resources such as photovoltaic solar panels is a very clean resource and represents a strong alternative to fossil fuels, which can harm the environment and cause pollution in the area. The selection of solar panels was based on studies we conducted, which found that the high solar irradiance and climatic conditions in Kuwait make solar energy a suitable and highly effective solution for generating electricity in the area. Through this project, the system is designed to cover approximately 50% of the residents' electricity requirements, demonstrating the project's success when implemented as planned.

In addition, the project highlighted its environmental benefits in reducing greenhouse gas emissions, as well as its positive social and economic impact, including raising public awareness, creating job opportunities, and achieving long-term cost savings. The Human Development Index assessment indicated that the transition to renewable energy will contribute to comprehensive sustainable development. Overall, the results of this project study indicate that implementing photovoltaic solar energy technology in residential areas such as Al-Khaldiya will play a key role in reducing the use of fossil fuels and achieving a more sustainable and resilient energy future.

11. Appendix : Interview Transcript with Dr. Morsi Ahmed

Date: November 3, 2025

Time: 12:00 PM

Type: Recorded Interview

Interviewee: Dr. Morsi Ahmed

Position: Head of the Material Science and Engineering Department, Chair of the Environmental Engineering Program, Abdullah Salem University

Interviewers : Retaj Bader , Shahed Yousef and Nada Ibrahim

[Recording Begins]

Interviewer:

Good afternoon, Dr. Morsi. It's a pleasure to meet you.

Dr. Morsi Ahmed:

Nice meeting you too.

Interviewer:

Before we start, could you please introduce yourself?

Dr. Morsi Ahmed:

Of course. I'm Dr. Morsi Ahmed, the Head of the Material Science and Engineering Department. I'm also serving as the Chair of the Environmental Engineering Program here at Abdullah Salem University. I graduated from Virginia Tech and spent 7 years there. After that, I moved to Germany, where I worked at the Karlsruhe Institute of Technology for another 7 years. Then I joined King Fahd University of Petroleum and Minerals for 7 years. Finally, in 2024, I moved here to Abdullah Salem University.

Interviewer: That's an impressive background, Dr. Morsi. Thank you. My first question is: In your opinion, what are the main challenges that countries face when trying to shift from traditional energy sources to renewable ones?

Dr. Morsi Ahmed:

That's a very good and broad question. I'll try to summarize it. There are several challenges, especially in the Gulf region. The most important challenge is infrastructure. All the infrastructure in the Gulf area is designed for traditional oil and gas. Moving from oil and gas to renewable energy requires large investments and new infrastructure, which affects the economic side. The second challenge is technical. The Gulf region faces dust and high temperatures, both of which reduce the efficiency of renewable energy systems such as solar panels. Dust accumulates on panels and decreases performance, while high temperature also lowers their efficiency. Another challenge is cost. Currently, the prices of renewable energy are still not as competitive as traditional energy sources. So, to summarize the main challenges are: infrastructure and economic factors, technical limitations (dust and high temperatures), and cost competitiveness.

Interviewer:

That's very clear, thank you. My second question is: What kind of materials are the most promising for improving renewable energy technologies, like solar panels or wind turbines?

Dr. Morsi Ahmed:

From a materials science point of view, there are several promising materials. First, Perovskites these materials can significantly increase the efficiency of solar panels. Second, carbon-based composites these materials are lightweight and resistant to corrosion, making them ideal for wind energy systems. Finally, MOFs, or Metal-Organic Frameworks. This material was invented by a Nobel laureate, Omar Yaghi, at the University of California, Berkeley. MOFs can store hydrogen, which makes them very important for green hydrogen production and the future of renewable energy. So, these three materials Perovskites, carbon-based composites, and MOFs are strong candidates for improving renewable energy technologies.

Interviewer:

That's fascinating, Dr. Morsi. Thank you very much for your time and insightful answers.

Dr. Morsi Ahmed: You're welcome. It was my pleasure.

[Recording Ends]

12. REFERENCES

Central Statistical Bureau of Kuwait. (n.d.). Population statistics.

<https://www.csb.gov.kw>

International Energy Agency (IEA). (n.d.). Kuwait energy profile.

<https://www.iea.org/countries/kuwait>

Munoz, J., Roberts, L., & Chen, T. (2021). Public health benefits of renewable energy transitions.

National Research Council. (2010). Hidden costs of energy: Unpriced consequences of energy production and use. The National Academies Press. <https://doi.org/10.17226/12794>

U.S. Environmental Protection Agency. (2023). Greenhouse gas equivalencies calculator.

<https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator>

Ahmed, M. (2025, November 3). *Recorded interview on renewable energy challenges and materials*. Conducted by R.

Bader, S. Yousef, & N. Ibrahim, Abdullah Salem University. (Personal communication, not publicly available).

International Renewable Energy Agency. (2023). *Renewable energy market analysis: GCC region*. IRENA.

Kuwait Central Statistical Bureau. (2024). *Statistical review of population, income, and development indicators*. Kuwait Government.

Ministry of Electricity, Water and Renewable Energy. (2024). *Annual statistical report*. State of Kuwait.

United Nations Development Programme. (2024). *Human development report 2024*. UNDP. <https://hdr.undp.org>

World Bank. (2024). *World development indicators: Kuwait*. World Bank Group.

World Health Organization. (2023). *Air pollution and public health impacts in the Eastern Mediterranean region*. WHO.

Wikipedia contributors. (2022). *Khaldiya (Kuwait)*. Wikipedia. Retrieved April 27, 2026, from

[https://en.wikipedia.org/wiki/Khaldiya_\(Kuwait\)](https://en.wikipedia.org/wiki/Khaldiya_(Kuwait))

Al-Shammari, M., & Al-Mahdi, A. (2022). Solar irradiance assessment for urban districts in Kuwait. (Note: Replace with actual journal/article title if known; this is a placeholder as the original text provided only the authors and year.)

National Renewable Energy Laboratory. (2023). PVWatts v8 calculator. U.S. Department of Energy.

<https://pvwatts.nrel.gov/>

Course Project Guidelines. (2025). Sustainable energy system for a Kuwaiti district.